Unit 28: Indefinite Integrals and the Substitution Method — Full Solutions

Warm-up

1. $\int (x^2+1)^3(2x)\,dx$

Let $u = x^2+1$, so $du = 2x\,dx$ — this matches exactly what's left over.

$$\int u^3\,du = \frac{u^4}{4} + C = \frac{(x^2+1)^4}{4} + C$$

2. $\int (3x+1)^4\,dx$

Let $u = 3x+1$, so $du = 3\,dx \Rightarrow dx = \frac{du}{3}$.

$$\int u^4 \cdot \frac{du}{3} = \frac{1}{3} \cdot \frac{u^5}{5} + C = \frac{u^5}{15} + C = \frac{(3x+1)^5}{15} + C$$

3. $\int 2x\sqrt{x^2+4}\,dx$

Let $u = x^2+4$, so $du = 2x\,dx$.

$$\int \sqrt{u}\,du = \int u^{1/2}\,du = \frac{2}{3}u^{3/2} + C = \frac{2}{3}(x^2+4)^{3/2} + C$$

4. $\int (x^3+2)^2(3x^2)\,dx$

Let $u = x^3+2$, so $du = 3x^2\,dx$.

$$\int u^2\,du = \frac{u^3}{3} + C = \frac{(x^3+2)^3}{3} + C$$

5. $\int (2x-5)^6\,dx$

Let $u = 2x-5$, so $du = 2\,dx \Rightarrow dx = \frac{du}{2}$.

$$\int u^6 \cdot \frac{du}{2} = \frac{1}{2} \cdot \frac{u^7}{7} + C = \frac{u^7}{14} + C = \frac{(2x-5)^7}{14} + C$$

6. $\int x(x^2+3)^5\,dx$

Let $u = x^2+3$, so $du = 2x\,dx \Rightarrow x\,dx = \frac{du}{2}$.

$$\int u^5 \cdot \frac{du}{2} = \frac{1}{2} \cdot \frac{u^6}{6} + C = \frac{u^6}{12} + C = \frac{(x^2+3)^6}{12} + C$$

7. $\int \sin(3x)\,dx$

Let $u = 3x$, so $du = 3\,dx \Rightarrow dx = \frac{du}{3}$.

$$\int \sin u \cdot \frac{du}{3} = \frac{1}{3}(-\cos u) + C = -\frac{1}{3}\cos(3x) + C$$

Standard

8. $\int \frac{4x}{\sqrt{2x^2+1}}\,dx$

Let $u = 2x^2+1$, so $du = 4x\,dx$ — matches the numerator exactly.

$$\int \frac{1}{\sqrt{u}}\,du = \int u^{-1/2}\,du = 2u^{1/2} + C = 2\sqrt{2x^2+1} + C$$

9. $\int x^2(x^3+1)^4\,dx$

Let $u = x^3+1$, so $du = 3x^2\,dx \Rightarrow x^2\,dx = \frac{du}{3}$.

$$\int u^4 \cdot \frac{du}{3} = \frac{1}{3} \cdot \frac{u^5}{5} + C = \frac{u^5}{15} + C = \frac{(x^3+1)^5}{15} + C$$

10. $\int \cos(2x+1)\,dx$

Let $u = 2x+1$, so $du = 2\,dx \Rightarrow dx = \frac{du}{2}$.

$$\int \cos u \cdot \frac{du}{2} = \frac{1}{2}\sin u + C = \frac{1}{2}\sin(2x+1) + C$$

11. $\int \frac{2x}{(x^2+1)^2}\,dx$

Let $u = x^2+1$, so $du = 2x\,dx$.

$$\int u^{-2}\,du = \frac{u^{-1}}{-1} + C = -\frac{1}{u} + C = -\frac{1}{x^2+1} + C$$

12. $\int \sec^2 x \tan x\,dx$

Let $u = \tan x$, so $du = \sec^2 x\,dx$.

$$\int u\,du = \frac{u^2}{2} + C = \frac{\tan^2 x}{2} + C$$

13. $\displaystyle\int (2x+3)(x^2+3x+1)^2\,dx$

Let $u = x^2+3x+1$, so $du = (2x+3)\,dx$ — matches exactly.

$$\int u^2\,du = \frac{u^3}{3} + C = \frac{(x^2+3x+1)^3}{3} + C$$

14. $\displaystyle\int 3x^2 \sin(x^3)\,dx$

Let $u = x^3$, so $du = 3x^2\,dx$.

$$\int \sin u\,du = -\cos u + C = -\cos(x^3) + C$$

Challenge

15. $\displaystyle\int x\sqrt{x+1}\,dx$

Let $u = x+1$, so $du = dx$. This also means $x = u-1$ — we need this to replace the lone x still sitting in the integral.

$$\int (u-1)\sqrt{u}\,du = \int (u-1)u^{1/2}\,du = \int \left(u^{3/2} - u^{1/2}\right) du$$

$$= \frac{2}{5}u^{5/2} - \frac{2}{3}u^{3/2} + C$$

Substitute back $u = x+1$:

$$= \frac{2}{5}(x+1)^{5/2} - \frac{2}{3}(x+1)^{3/2} + C$$

16. $\displaystyle\int_0^1 2x(x^2+1)^3\,dx$

Let $u = x^2+1$, so $du = 2x\,dx$. Change the bounds: when $x=0$, $u=1$; when $x=1$, $u=2$.

$$\int_1^2 u^3\,du = \left[\frac{u^4}{4}\right]_1^2 = \frac{16}{4} - \frac{1}{4} = 4 - 0.25 = 3.75 = \frac{15}{4}$$

Answer: $\dfrac{15}{4}$.

17. $\displaystyle\int_0^2 \frac{x}{\sqrt{x^2+9}}\,dx$

Let $u = x^2+9$, so $du = 2x\,dx \Rightarrow x\,dx = \dfrac{du}{2}$. Change the bounds: when $x=0$, $u=9$; when $x=2$, $u=13$.

$$\int_9^{13} \frac{1}{\sqrt{u}}\cdot\frac{du}{2} = \frac{1}{2}\int_9^{13} u^{-1/2}\,du = \frac{1}{2}\Big[2\sqrt{u}\Big]_9^{13} = \Big[\sqrt{u}\Big]_9^{13} = \sqrt{13}-\sqrt{9} = \sqrt{13}-3$$

Answer: $\sqrt{13}-3 \approx 0.606$.

18. $\displaystyle\int \sin^3 x\cos x\,dx$

Let $u = \sin x$, so $du = \cos x\,dx$.

$$\int u^3\,du = \frac{u^4}{4}+C = \frac{\sin^4 x}{4}+C$$

19. $\displaystyle\int x^3\sqrt{x^4+5}\,dx$

Let $u = x^4+5$, so $du = 4x^3\,dx \Rightarrow x^3\,dx = \dfrac{du}{4}$.

$$\int \sqrt{u}\cdot\frac{du}{4} = \frac{1}{4}\cdot\frac{2}{3}u^{3/2}+C = \frac{1}{6}u^{3/2}+C = \frac{1}{6}(x^4+5)^{3/2}+C$$

Applied

20. $\dfrac{dP}{dt} = \dfrac{100t}{\sqrt{t^2+9}}$

Let $u = t^2+9$, so $du = 2t\,dt \Rightarrow t\,dt = \dfrac{du}{2}$.

$$P(t) = \int \frac{100t}{\sqrt{t^2+9}}\,dt = 100\int \frac{1}{\sqrt{u}}\cdot\frac{du}{2} = 50\int u^{-1/2}\,du = 50(2u^{1/2})+C = 100\sqrt{u}+C$$

Answer: $P(t) = 100\sqrt{t^2+9}+C$.

21. $MR(x) = 6x^2(x^3+10)^2$

Let $u = x^3+10$, so $du = 3x^2\,dx \Rightarrow x^2\,dx = \dfrac{du}{3}$.

$$R(x) = \int 6x^2(x^3+10)^2\,dx = 6\int u^2\cdot\frac{du}{3} = 2\int u^2\,du = 2\cdot\frac{u^3}{3}+C = \frac{2}{3}u^3+C$$

Answer: $R(x) = \dfrac{2}{3}(x^3+10)^3 + C$.

22. $\displaystyle\int_0^2 \frac{8t}{(t^2+1)^2}\,dt$

Let $u = t^2+1$, so $du = 2t\,dt \Rightarrow t\,dt = \dfrac{du}{2}$. Change the bounds: when $t=0$, $u=1$; when $t=2$, $u=5$.

$$8\int_1^5 \frac{1}{u^2}\cdot\frac{du}{2} = 4\int_1^5 u^{-2}\,du = 4\Big[-u^{-1}\Big]_1^5 = 4\left(-\frac{1}{5}-(-1)\right) = 4\left(\frac{4}{5}\right) = \frac{16}{5}$$

Answer: total water collected $= \dfrac{16}{5} = 3.2$ gallons.

23. $\displaystyle\int_0^{\sqrt{\pi}} t\sin(t^2)\,dt$

Let $u = t^2$, so $du = 2t\,dt \Rightarrow t\,dt = \dfrac{du}{2}$. Change the bounds: when $t=0$, $u=0$; when $t=\sqrt{\pi}$, $u=\pi$.

$$\int_0^{\pi} \sin u\cdot\frac{du}{2} = \frac{1}{2}\Big[-\cos u\Big]_0^{\pi} = \frac{1}{2}[(-\cos\pi)-(-\cos 0)] = \frac{1}{2}[1-(-1)] = \frac{1}{2}(2) = 1$$

Answer: exact distance traveled $= 1$ meter.